

ICAI CAPSTONE PROJECT

(AICA-L2-B83 - SURAT - 15,16 AUG 2026 & 19,20,21 AUG 2026)

OFFLINE TOOL FOR PREPARATION OF QUARTERLY TDS RETURN

Automating Form 140 (New 26Q) under the Income-tax Act, 2025 • Fully offline
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1. The Problem (our day-to-day pain)

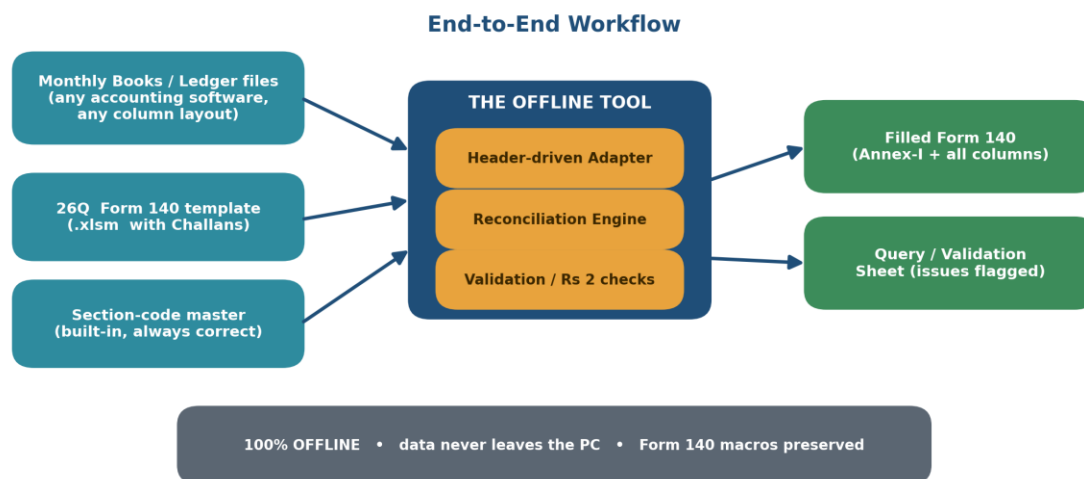
- Every client's software exports books data differently — column names, order and layout keep changing, so a fixed template never fits.
- Deductees must be matched to the right challan and section — done by hand, month by month, and easy to get wrong.
- Form 140 needs exact coded values (e.g. section “1006-194H”, deductee “01-Companies”) or the return is rejected.
- Small Amount × Rate vs TDS-deducted mismatches slip through and surface only at validation.
- From FY 2026-27 the format changed to Form 140 with new section numbering — everyone must relearn it.
- Client data is confidential — online converters are not acceptable.

2. Objectives

- Remove manual re-keying of deductee data into the return template.
- Work with any books layout, reconcile to challans automatically, and flag — never hide — differences.
- Fill every statutory column in the exact coded format; catch errors before filing; stay 100% offline.
- Package as a simple .exe any colleague can run with nothing installed.

3. The Solution — an Offline TDS Return Tool

- Three inputs → two outputs, entirely on the local PC:



4. Key Features

- **Reads any layout:** a header-driven “adapter” matches columns by name (tolerates renames, re-ordering, typos).
- **Auto challan reconciliation:** ties each deductee to the correct challan; totals checked to the rupee; gaps reported, not forced.
- **Correct codes, always:** section as “1006-194H”, deductee as “01-Companies / 02-Other than Companies”, from a built-in master.
- **Fills every column:** challan serial, section, deductee code, PAN, name, amount, dates, rate and tax in Annex-I.
- **Built-in validations:** PAN format, out-of-quarter dates, duplicates, and a ₹2 check that Amount × Rate matches TDS deducted.
- **Colour-coded Query Sheet:** every issue by severity (High / Medium / Low) so review is fast.
- **Safe & genuine:** 100% offline; preserves the template's macros and dropdown lists.
- **One-click .exe:** runs on any Windows PC — no Python, no libraries, no internet.

5. How It Was Built

- Hand-mapped one real quarter to learn the exact Form 140 structure, then generalised it into a reusable adapter.
- Hardened against three genuinely different real export formats until all parsed with full accuracy.

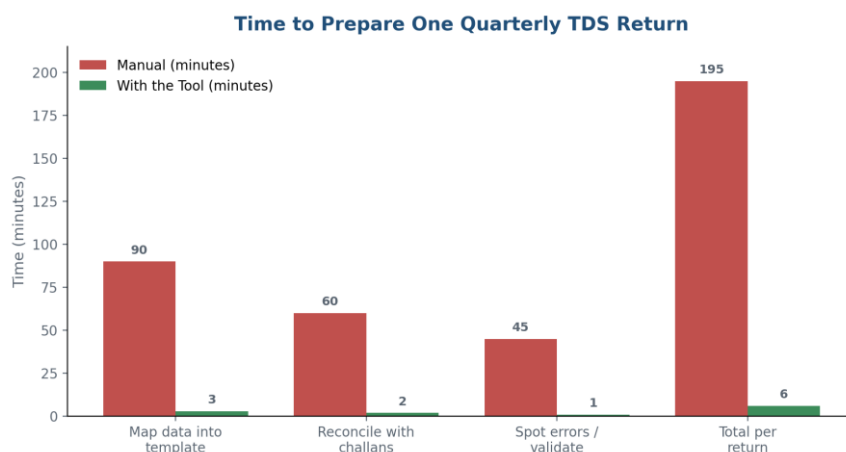
- Embedded the official section-code master; added the reconciliation + validation layers; built a click-based window.
- **Tech:** Python + openpyxl (Excel, macro-safe), Tkinter (window), PyInstaller (.exe). No cloud, no internet.

6. Challenges Solved

- Different column layouts every time → **header-driven fuzzy matching instead of fixed positions.**
- Same heading meant different things → **decide by the column's values, not just its name.**
- A grand-total row read as data → **guards skip subtotal / total / header rows.**
- Codes must be exact → **built-in official master always outputs the correct coded form.**
- Books vs challans didn't tie out → **report the exact rupee gap, never force a match.**
- Missing library / large .exe on a user PC → **auto-install on first run; excluded unused libraries and compressed the build.**

7. Time & Efficiency Gains

- **~3.15 hours per return → under 10 minutes** (illustrative), with higher accuracy as mapping/reconciliation errors are designed out.



8. ICAI Value, Future Scope & Conclusion

- **Skills shown:** applied TDS/Form 140 knowledge, automation & controls (audit trail via Query Sheet), data confidentiality, practical tech.
- **Future scope:** direct NSDL FVU file output; Forms 138 (Salary) & 143 (TCS); PAN-status check; multi-client dashboard.
- **Conclusion:** an everyday frustration turned into a dependable offline tool — minutes instead of hours, fewer errors, client data protected, and time freed for review and advisory.